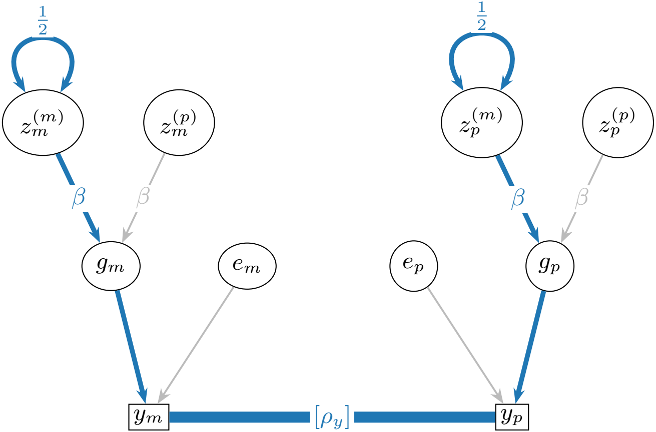

$$z_m^{(m)} \leftrightarrow z_m^{(m)} \rightarrow g_m \rightarrow y_m — y_p \leftarrow g_p \leftarrow z_p^{(m)} \leftrightarrow z_p^{(m)}$$
$$\mathrm{Cov}\left[z_m^{(m)}, z_p^{(m)}\right] = \frac{1}{2} \cdot \beta \cdot \frac{\rho_y}{V_E+\beta^2} \cdot \beta \cdot \frac{1}{2} = \frac{\beta^2 \rho_y}{4\left(V_E+\beta^2\right)}$$